ME 239: Introduction to Data Science for Mechanical Engineers Spring 2026 Syllabus

Course Information

- **ME 239: Introduction to Data Science for Mechanical Engineers**
- **CRN & Meeting Times**
    - **Section 003** – Taghian: M 12:30p-1:20p, ME G061
    - **Section 004** – Taghian: W 12:30p-1:20p, MEG061
    - **Section 005** – Maulik: F 12:30p-1:20p, MEG061
    - **Section 006 –** Mirfarah: T 2:30p-3:20p, MEG061
    - **Section 007** – Taghian: M 1:30p-2:20p, ME2061
    - **Section 008** – Taghian: W 1:30p-2:20p, ME2061
    - **Section 009** – Maulik: F 1:30p-2:20p, ME2061
    - **Section 010** – Mirfarah: T 8:30a-9:20a, WALC 1121
    - **Section 011** – Maulik: Th 8:30a-9:20a, WALC 1121
    - **Section 012** – Yang: W 3:30p-4:20p, Lecture Hall 105 (Indianapolis)
    - **Section 013** – Yang: M 3:30p-4:20p, Lecture Hall 105 (Indianapolis)
- **Instructional Modality: Face to Face**
- **Course credit hours: 1**

Instructor(s) Contact Information

Romit Maulik
- **Email:** rmaulik@purdue.edu
- **Office:**

Shengfeng Yang
- **Email:** shengfengyang@purdue.edu
- **Office: SL 160G (Indianapolis)**

Soudabeh Taghian Dinani
- **Email:** staghian@purdue.edu
- **Office: ARMS 1212**

Motahareh Mirfarah
- **Email:** mmirfara@purdue.edu

- **Office: FLEX 3081.14**

TA: Mohan Chi

- **Email:** chi70@purdue.edu

TA: Ashwin Suriyanarayanan

- **Email:** asuriyan@purdue.edu

Office Hours

In-person office hours will be held in the ME 239 Tutorial Room (ME 2134), and online office hours will be available via Zoom at this link: ME 239 Virtual Office Hours.

- **Maulik:**
  - In-person: Fridays, 2-3pm
- **Yang:**
  - In-person: Wednesdays, 2:30 pm -3:30 pm
- **Taghian:**
  - In-person: Wednesdays, 10:00 am -12:00 pm
- **Mirfarah:**
  - In-person: Tuesdays, 9:30 am-10:30 am
- **Chi:**
  - In-person: Wednesdays, 2:30 pm - 6:30 pm
  - Online: Thursday, 2:30 p.m. - 6:30, ME 239 Virtual Office Hours
- **Ashwin:**
  - In-person: Fridays, 2:30 pm – 6:30 pm
  - Online: Tuesdays, 3:30 pm – 7:30 pm, ME 239 Virtual Office Hours

Audience

Mechanical engineering sophomores

Course Description

This course introduces data science to sophomore mechanical engineers with no prior knowledge on the topic. We start by defining data science and demonstrating why mechanical engineers should care. Most data science nowadays happens in Python. So, the course starts with a brief overview of Python basics going from simple mathematical expressions to data loading and visualization. Data science must deal with uncertainty and randomness, and that is why we introduce several concepts from probability theory. We use these probabilistic concepts to summarize and compare datasets. Data science is about making models. For example, how can you make a model that will give you the efficiency of an engine given the operating conditions using experimental data? So, we end by showing how to use data to make models, and how to test if these models are good.

Learning Outcomes

After completing this course, you will be able to:

- Program in Python within a Jupyter notebook environment.
- Summarize and compare datasets using empirically estimated statistics.
- Summarize and compare datasets visually.
- Represent uncertainty using probabilities.
- Apply simple probability rules to propagate uncertainty.
- Estimate probabilities from data.
- Solve regression problems (learn from data a linear model that takes you from a set of input variables to a continuous variables).
- Solve classification problems (learn from data a model that takes you from a set of input variables to a discrete label).

Required Texts and Tools

Online Textbook

The online textbook can be found at this link.

Jupyter Notebook

Jupyter notebooks are interactive documents that can simultaneously contain text, mathematics, images, and executable code. The executable code can be in many programming languages (e.g., R, Matlab), but we are only going to use Python in this course. The course uses Jupyter notebooks for the following content: Reading Activities, Hands-on Activities, and Homework Assignments. The rationale behind this choice is that it allows the student to focus on the mathematical methods rather than the programming and it ensures the reproducibility of the course content. Of course, understanding the code in Jupyter notebooks does require knowledge of Python, albeit it does not require knowing how to structure and call Python code from the command line. Jupyter notebooks can be run either on the students' personal computers (instructions vary with operating system and can be found here) or in several cloud computing resources. The recommended method for this class is to use Google Colab which is available free of charge and requires only a standard Google account. The activity links included in the course will take you automatically to a copy of the latest version of the corresponding Jupyter notebook which you can then save and edit on your Google Drive.

Access to the Jupyter Notebooks Repository

As stated earlier,the recommended method for using the Jupyter notebooks of this class is to use Google Colab. The links to all the activities will take you directly to a Google Colab copy of the Jupyter notebook. If you want to use any alternative method (e.g., your personal computer or anything else), you will need access to the Jupyter Notebook repository for the class. Select this link for a Git version control repository.(If you have no idea what Git is, this is a good tutorial.)

Prerequisites

- Basic calculus.
- Matrix-vector multiplication.
- Some programming experience (e.g., Matlab or Python). You will learn Python as part of this course.

Course Schedule

Week	Topic	Reading	Homework	Deadline	Quiz
1	Introduction to data science, Python basics	Lecture 1	Homework 1	01/23/2026	
2	Data arrays, numerical Python	Lecture 2	Homework 2	01/30/2026	
3	Data loading and selection	Lecture 3	Homework 3	02/06/2026	
4	Data visualization	Lecture 4	Homework 4	02/13/2026	
5	Functions, data manipulation, and models	Lecture 5	Homework 5	02/20/2026	
6	Conditionals and loops	Lecture 6	Homework 6	02/27/2026	Quiz 1 (on HW1-4)
7	Probability as a measure of uncertainty	Lecture 7	Homework 7	03/06/2026	
8	The basic rules of probability	Lecture 8	Homework 8	03/13/2026	Quiz 2 (on HW5-6)
9	Discrete random variables	Lecture 9	Homework 9	03/27/2026	
10	Continuous random variables	Lecture 10	Homework 10	04/03/2026	
11	Expectation and variance	Lecture 11	Homework 11	04/10/2026	
12	Normal distribution, quantiles, credible intervals	Lecture 12	Homework 12	04/17/2026	Quiz 3 (on HW7-10)
13	Fitting models with the principle of maximum likelihood	Lecture 13	Homework 13	04/24/2026	
14	Covariance, correlation, and linear regression	Lecture 14	Homework 14	05/01/2026	
15	Linear regression continued	Lecture 15	Homework 15	05/08/2026	Quiz 4 (on HW11-13)
16	Classification via logistic regression	Lecture 16	Homework 16		

Grading

This course will be graded based on the following criteria:

Assessment Type	Description	% of Final Grade
Homework	There will be sixteen (16) homework assignments. The homework will be turned in on Gradescope by the assigned date at 11:59pm. The lowest homework will be dropped.	85%

Quizzes	The quizzes will be given in class and turned in on Gradescope (except for the first two, which are on Brightspace—see the Quizzes section for more details). These are to check your understanding of the homework material.	10%
Class Participation	Class participation grade will be at the discretion of the section instructor.	5%

Homework Assignment Lateness and Regrade Policy – Assignments will be accepted **only** upto one day later for which we will apply a **10% late penalty**. **Regrade requests** will be accepted **within one week** after the grades for each assignment or quiz are released.

Grading Scale

Your course grade will be based on the following grading scale:

A+ = > 98%	B+ = 88-89%	C+ = 78-79%	D+ = 68-69%	F < 60%
A = 92-97%	B = 82–87%	C = 72-77%	D = 62-67%	
A- = 90–91%	B- = 80-81%	C- = 70-71%	D- = 60-61%	

The instructor reserves the right to change the grading scale in a way that favors students.

Homework

The homework will be turned in on Gradescope by Friday each week at 11:59pm.

Quizzes

The first two quizzes will be available on Brightspace (Tools → Quizzes). One will cover academic integrity (you will only acknowledge that you have read it and will follow it), and the other will cover the syllabus (you will only type your name as an acknowledgment that you have read the syllabus carefully and will follow the guidelines in it).

The remaining (pen-on-paper) quizzes will be given in class. Each quiz will be completed in the last 15 to 20 minutes of class (depending on its content). Please check course schedule for details about their time and materials each covers. During the quiz, the use of cell phones is not allowed. You will need a simple calculator for some of the quizzes. Please make sure to bring one to every class session, as the quizzes are closed-book and you are not allowed to use your phone or laptop during quizzes.

Getting help

Office Hours – Office hours are mentioned above. Please refer to Brightspace for the link and review the guidelines for attending office hours.

Campuswire – We will use Campuswire as the asynchronous discussion board. Please check Brightspace for the link and review the guidelines for posting.

Email – If you have a course-related question that is not personal or private, please post it on Piazza (you can post anonymously if you wish) or join our in-person/online office hours. For personal or private questions, contact the instructor of your section. Even if you contact a TA, always CC the course instructor.

Note – For all questions posted on Piazza or via email, allow up to one business day (excluding holidays) for a response. For example, if you post on Friday, you should expect a response by Monday. Plan ahead and ask questions early to ensure you get an answer on time. Also, please note that although assignments are due by Friday at midnight, the last time you can expect a response from us is 5 PM on Fridays (or within one business day). Questions asked after 5 PM may not receive a response before the deadline, so please plan accordingly.

Discussion Guidelines

Please follow the Discussion Guidelines when contributing to discussions in this course. Here are a few of the key points you should remember:

- Do not use offensive language. Present ideas appropriately.
- Be cautious in using Internet language. For example, do not capitalize all letters since this suggests shouting.
- Avoid using vernacular or slang language. This could possibly lead to misinterpretation.
- Do not hesitate to ask for feedback.
- Be concise and to the point.
- Think and edit before you push the "Send" button.

Generative AI Policy

Use of AI, Large Language Models, and Generative Tools

You may use AI tools in this course, but **only** in an ethical way and as a learning aid, not as a replacement for your own work. AI should support your understanding of concepts—not do the assignments for you. Below are appropriate and inappropriate uses of AI.

Appropriate Uses of AI

AI tools (e.g., ChatGPT, Google Gemini) can be used effectively for the following purposes:

- **As a Tutor:** Ask AI to explain a concept you don't understand (e.g., the input() function — in this link, I used it for this purpose).
- **Debugging Support:** Use AI to help identify and fix bugs in your code (in this link, I used it for this purpose).
- **Code/Concept Clarification:** Ask AI to explain specific lines of code from lecture, the textbook, or online resources.
- **Code Improvement:** Request suggestions for improving or optimizing code you wrote (in this link, I used it for this purpose).

- **Text Improvement / Paraphrasing:** You may use AI to improve the clarity, grammar, or phrasing of explanations, comments, or documentation. *(For example, this section of the syllabus was improved using AI.)*

When you use ChatGPT, Gemini, or any other AI tool, you **must cite it clearly**. For example, you should state in your code comments how you used the tool and provide a link to the AI conversation.

Examples of correct citation:

- *"Used ChatGPT to understand the input() function at line 12 (link)."*
- *"Used Gemini for debugging suggestions on lines 30–35 (link)."*
- *"Asked ChatGPT for ideas to improve efficiency of the loop at line 50 (link)."*

Note:

- You must cite all references you used, including AI tools, textbooks, websites, or other resources. You can do this in one of two ways: Add the citation next to the relevant code in comments, OR List the citation at the beginning of your code (after the academic integrity statement), clearly mentioning which line numbers it applies to.
- Even with AI assistance, the work you submit must be your own and you must understand every line of it.
- The code you submit must reflect your own understanding. You must be able to explain every line of your code.
- All submitted code must include line-by-line comments showing your understanding.
- If we suspect you used AI inappropriately, you may be asked to explain your code. If you cannot explain it, consequences may include: receiving a zero on the assignment, a failing grade in the course, and/or a report to ODOS for an academic integrity violation.

Incorrect Uses of AI (Not Allowed)

- Asking AI to solve the entire assignment by Copying and pasting assignment instructions, quizzes, or screenshots into AI to get full answers.
- Using code-completion/autofill tools like GitHub Copilot or Gemini's code autocompletion. These tools generate entire blocks of code without requiring your own reasoning and must be deactivated. Remember: just as no one would hire an accountant who needs a calculator to add simple numbers (e.g., 1 + 10), no one will hire a programmer who needs AI to write a basic loop or if statement.

Side Note: Academic Integrity Reminder

- At the start of every assignment, include an **academic integrity statement** in the comments.
- List any collaborators you discussed the assignment with (by name).
- You may discuss general approaches with classmates, but you must complete and write the code **individually**.

- You may not look at or share code with other students.
- If similarity checks reveal copied or shared solutions, all involved students will be reported to ODOS for an integrity violation.

Attendance Policy

This course follows Purdue's attendance regulations, which require students to be present for every scheduled class. Attendance will be taken at the start of class, and lateness will be noted.

If you know in advance that you will miss class—for example, for a University event or religious observance—you should inform your instructor as soon as possible. If an emergency prevents you from giving advance notice, contact your instructor by email as soon as you can.

If you are unable to reach your instructor directly due to circumstances beyond your control, you (or someone on your behalf) should notify the Office of the Dean of Students by contacting them or going to their website and complete their online forms. Excused absences may be granted for reasons such as illness, bereavement, military service, jury duty, or parenting leave. For more information, see the University Regulations website.

Class Expectations

You are expected to attend and participate in all class sessions unless excused as described above. Attendance may be checked more than once during class to confirm engagement throughout the period.

- **Preparation**: Come to class prepared, ready for quizzes, and able to take part in activities.
- **Punctuality**: Arrive on time.
    - Arriving up to 15 minutes late will be recorded as a lateness. Arriving more than 15 minutes late will be counted as an absence.
    - Leaving class early without prior notice and a valid excuse will also count as a lateness.
    - Three lateness equal one unexcused absence.
    - If you have travel-time concerns, discuss them with your instructor.
- **Attentiveness**: Being physically present is not enough—you are expected to pay attention and participate.
    - Laptops should only be used for class-related work.
    - Using a laptop for games or unrelated assignments will result in being marked as **distracted**, which counts as a lateness.

Accessibility Information

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247

- Google Accessibility
- Jupyter Notebook a11y toolbar (from Microsoft research)

Mental Health/Wellness Statement:

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack**.** Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours. The CAPS website also offers resources specific to situations such as COVID-19.

Academic Integrity

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing or by calling 765-494-8778. While information may be submitted anonymously, the more information that is submitted provides the greatest opportunity for the university to investigate the concern.

The Purdue Honor Pledge

"As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue"

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Link to Purdue's nondiscrimination policy statement.

Diversity and Inclusion Statement

In our discussions, structured and unstructured, we will explore a variety of challenging issues, which can help us enhance our understanding of different experiences and perspectives. This can be challenging,

but in overcoming these challenges we find the greatest rewards. While we will design guidelines as a group, everyone should remember the following points:

- We are all in the process of learning about others and their experiences. Please speak with me, anonymously if needed, if something has made you uncomfortable.
- Intention and impact are not always aligned, and we should respect the impact something may have on someone even if it was not the speaker's intention.
- We all come to the class with a variety of experiences and a range of expertise, we should respect these in others while critically examining them in ourselves.

Course Evaluation

At the end of the course, you will be provided with an opportunity to evaluate this course and your instructor. Your participation is an integral part of this course, and your feedback is vital to improving education at Purdue University. I strongly urge you to participate in the survey.

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

Here are the links to the building emergency plans:

- ME building emergency plan
- MSE building emergency plan

Disclaimer

This syllabus is subject to change.